Plotting Dyson Orbital

S. Gulania

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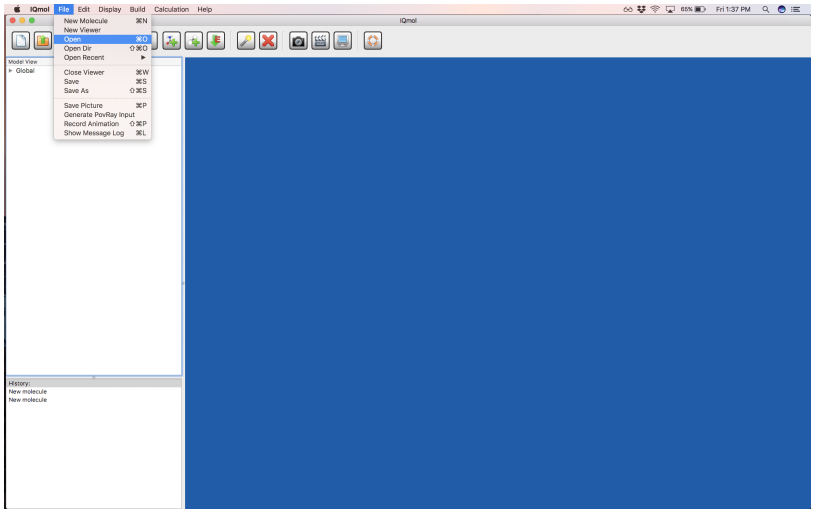
1 Procedure (Q-Chem 5.1 and later)

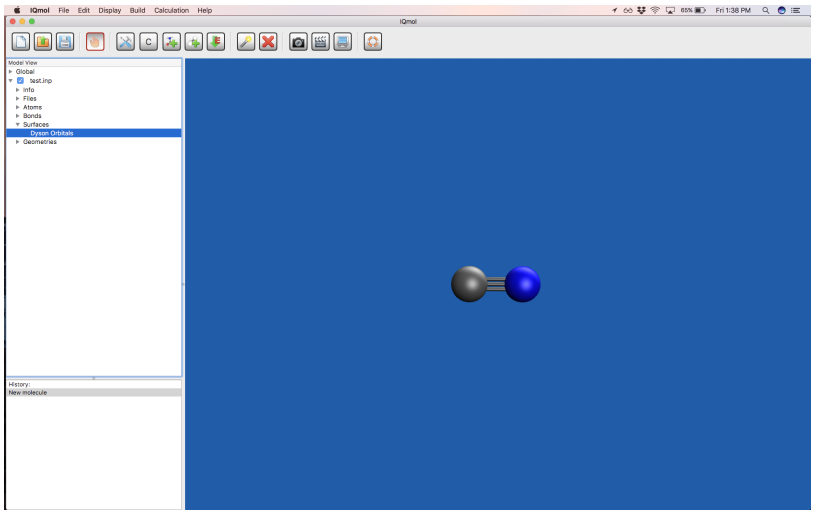
- Run the input for computing Dyson orbital according to Q-Chem manual. Important keywords are `PRINT_GENERAL_BASIS = TRUE`, `CC_DO_DYSON=TRUE`. Note - adding `GUI = 2` will produce .fchk file, which can also be used to plot Dyson orbital.

```
$comment
  Input for generating Dyson orbital using output file.
$end

$molecule
-1 1
 C         0.0000000000    0.0000000000   -0.6474385113
 N         0.0000000000    0.0000000000    0.5217097297
$end

$rem
   BASIS  = 6-31G*
   PRINT_GENERAL_BASIS = TRUE
   JOB_TYPE  =  SP
   METHOD  =  EOM-CCSD
   EOM_IP_ALPHA  =  [1,0,0,0]
   CC_DO_DYSON = TRUE
$end
```

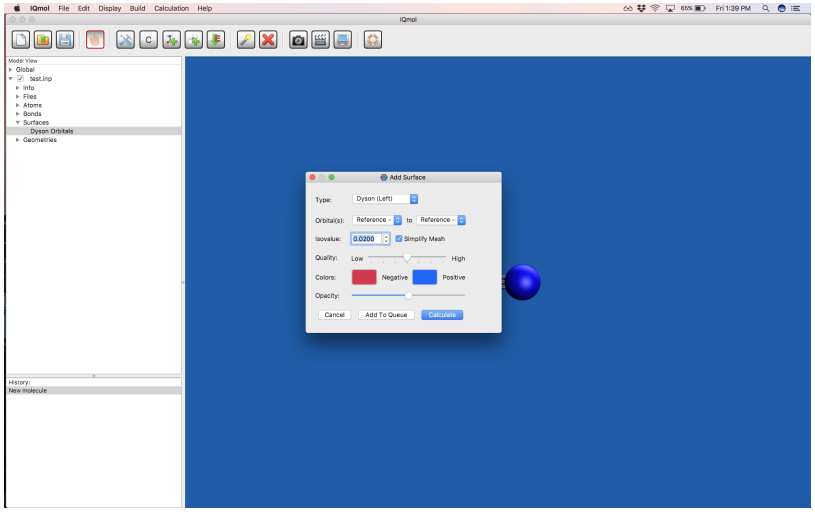
- Open `filename.inp.out` in IQmol.



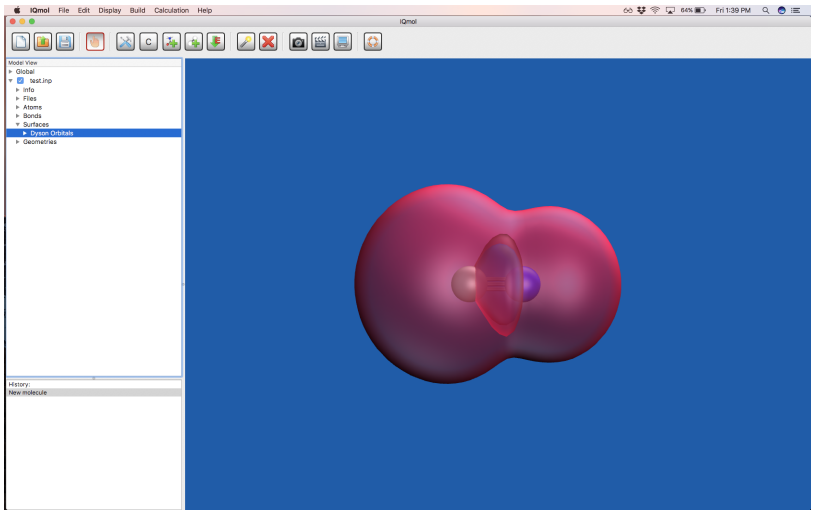
Open the output file



Select Dyson orbital from Surfaces option on the left side panel



Select the transition for Dyson orbital



Dyson orbital of CN^-

2 Procedure (using cube files)

- Important keywords are `CC_DO_DYSON=TRUE`, `IANLTY=200`, `MAKE_CUBE_FILES=TRUE` The `plots` box set up the grid for each orbitals at which the densities are generated.

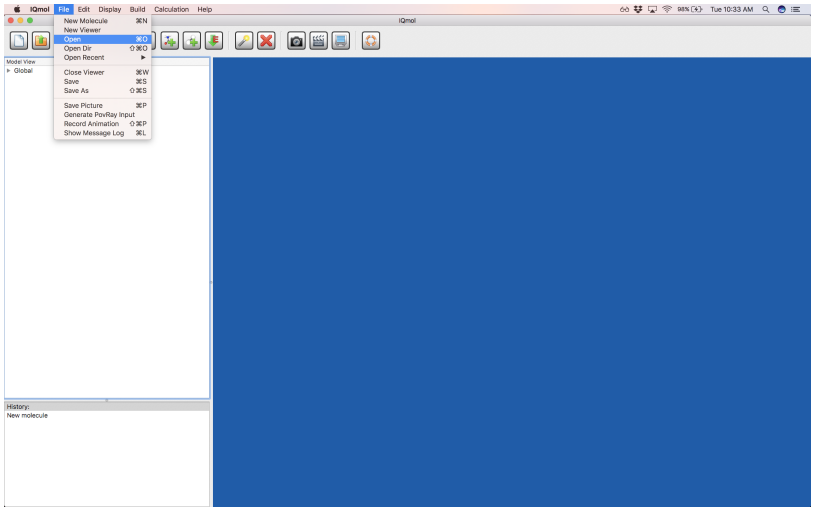
```
$comment
  Input for generating Dyson orbital using cube files
$end

$molecule
-1 1
 C         0.0000000000    0.0000000000   -0.6474385113
 N         0.0000000000    0.0000000000    0.5217097297
$end

$rem
   JOB_TYPE  =  SP
   BASIS  = 6-31G*
   METHOD  =  EOM-CCSD
   EOM_IP_ALPHA  =  [1,0,0,0]
   CC_DO_DYSON  = TRUE
   IANLTY   = 200
   MAKE_CUBE_FILES = TRUE
$end

$plots
plots Dyson orbital (setting up grid)
   30   -4   4
   30   -4   4
   30   -4   4
   0     0   0    0
$plots
```

- This will generate cube files in your working directory where the input exist with extension `filename.inp.out.plots`. If it does nopt generate use the $-save$ option mentioned in QChem 5.1 manual page 39

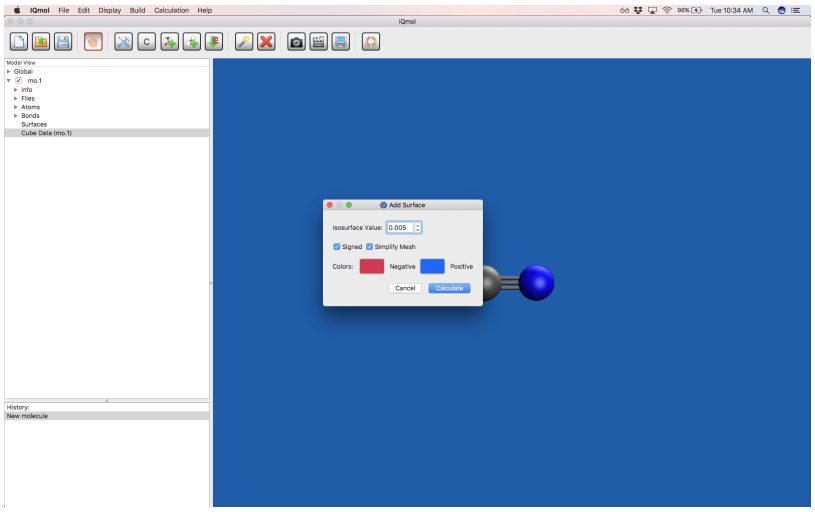
  (`http://hogwarts.usc.edu/~krylov/manual.pdf`)

- Now `filename.inp.out.plots` folder will have the respective cube files for each Dyson orbital (righ and left). The cube files can be loaded in IQmol (`http://iqmol.org`), which is an open-source software.

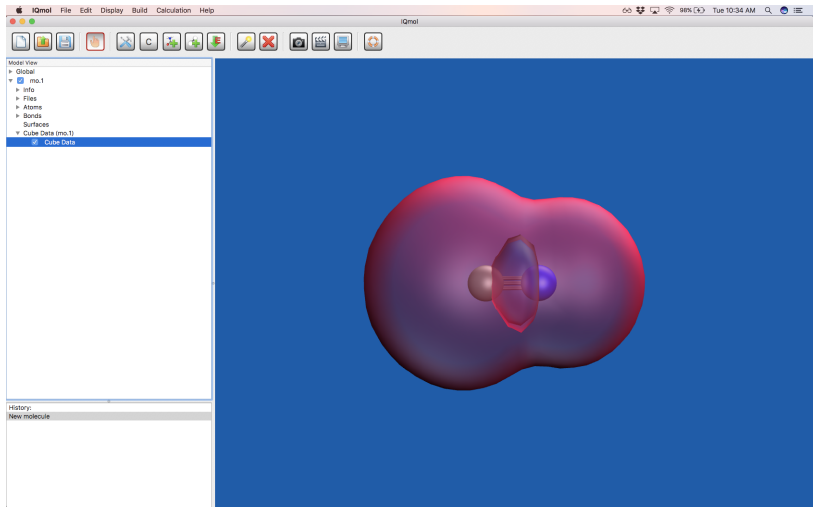
- Open IQmol and follow the instructions mentioned below



Open the cube file



Select Cube Data option from the left side panel



Dyson orbital of CN^-

3 Procedure (using .mo files)

Follow the instruction provided by Prof. C. David Sherrill at
`http://vergil.chemistry.gatech.edu/resources/plot-orbs.html`

4 Important notes

- Dyson orbitals are defined as the overlap between an N electron molecular wavefunction and the N-1 electron wave function of the ionized system (`http://iopenshell.usc.edu/downloads/ezdyson/`)
- ezDyson is an open-source code that calculates absolute photodetachment/ photoionization cross sections, photoelectron angular distributions (PADs), and anisotropy parameters (β) using Dyson orbitals computed by an ab initio program.